
The Flow of Information in European Book Publishing, 1500-1800

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Abstract

The origins of European innovativeness in the centuries leading up to the industrial revolution relies on the networks of information sharing across the competing polities (Mokyr 2016, Henrich 2020). Newly available digital datasets are allowing these connections to be studied quantitatively on a large scale, e.g. in scholarly networks (Edelstein et al. 2017, De La Croix et al. 2025). An interesting opportunity lies in monitoring the flow of information directly that has so far only been done in smaller case studies (Valleriani et al. 2022, Zapperi et al. 2025).

We adopted a novel dataset to track the spread of information directly on a large scale: the Heritage of Printed Books (HPB) database. HPB is an aggregation of historical book metadata from the catalogues of 60+ libraries, containing records on ca. 10 million editions from 1454–1830. Following bibliographic data science principles (Lahti et al. 2019), we harmonized and organized the data to study the diffusion of popular authors ($n = 16,574$) of printed books in different cities ($n = 1,486$) across Europe.

We measured the speed by which authors came to be published across different regions in Europe. Considering the diffusion of each author across cities as an information cascade, and using network-based diffusion analysis, we also inferred the latent networks between cities that would support these cascades (Franz & Nunn 2009, Gomez-Rodriguez et al. 2010), see Figure 1. We then explored the dependency of typical pathways on authors starting in different eras and regions. Across time periods, we see strong connections between current major publishing centers as well as regional attractors that act as stepping stones for smaller locations. Through patterns of book publication we can directly reconstruct the flow of information during a consequential time in European history.

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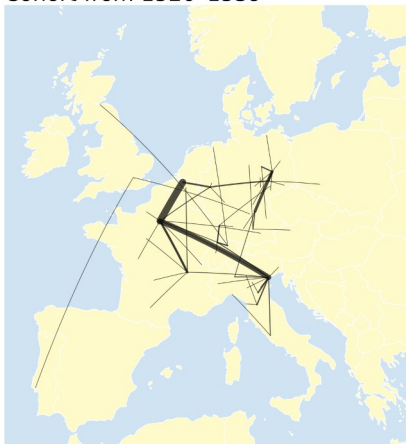
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Cohort from 1520-1539



Cohort from 1640-1659



Cohort from 1760-1779



Figure 1. Reconstructed network of information flow between cities for authors starting their careers in three cohorts 1520–1539, 1640–1659, 1760–1779. The network is built on the first 50 years of each author's career. Data: HPB